

Data Validation & Analysis Team Requirements

For Brain Tumor Classification & Segmentation Datasets

Purpose	Define the independent QA, clinical/label review, leakage detection, segmentation-mask checks, dataset analysis, and model-readiness criteria required before training.
Primary users	Data Validation & Analysis Team, AI Model Team, Medical Reviewers
Scope	MRI data for brain tumor classification and segmentation model development.

Project task definition
Classification classes: glioma, meningioma, pituitary tumor, and no tumor. Segmentation: pixel/voxel-level tumor masks; binary tumor-vs-background is the minimum acceptable target, with multi-region masks retained when available.

1. Team Mission and Authority

The Data Validation & Analysis Team independently decides whether collected data are suitable for model development. It must not assume that a public dataset is correct simply because it is popular or previously used.

- **Validate structure:** files, dimensions, metadata, image-mask pairing, readable formats, and manifest completeness.
- **Validate identity:** patient grouping, exact duplicates, near-duplicates, and cross-split leakage.
- **Validate labels:** class-label consistency and clinical plausibility through structured review.
- **Validate masks:** alignment, label values, geometry, completeness, and suspicious annotations.
- **Analyze bias:** class balance, source distribution, sequence/resolution differences, and split mismatch.
- **Decide status:** accept, reject, or escalate samples/sources for medical/manual review.

Critical independence rule

Collection counts are not final training counts. Only samples marked ACCEPTED after validation may enter the model-ready dataset.

2. Required Inputs from the Collection Team

Check	Requirement
☐	Raw or source-preserved image files.
☐	Segmentation masks in original format and their label definitions.
☐	Master manifest containing sample_id, subject_id where available, source, task, labels, modality, paths, and original split.
☐	Source registry with usage/license information and version/download date.
☐	Source documentation describing label construction and any preprocessing.
☐	Collection summary and exception log.

3. Validation Status Vocabulary

Status	Meaning	Training use
ACCEPTED	Passed all mandatory checks or approved after review	May enter model-ready dataset
REVIEW	Potential issue; manual/medical review required	Must not enter training yet
REJECTED	Broken, duplicated/leaky, wrong label/mask, unusable quality, or unacceptable provenance	Exclude
QUARANTINED	Source-level concern or unresolved metadata/usage issue	Keep separate until resolved

4. Validation Pipeline - Required Order

1. Manifest and file integrity validation.
2. Subject/patient identity grouping and duplicate/near-duplicate detection.
3. Classification label audit and class mapping verification.
4. Segmentation mask structural and geometric validation.
5. Image quality and artifact screening.
6. Dataset-level statistical analysis: classes, sources, modalities, dimensions, tumor sizes, and missingness.
7. Train/validation/test split construction at patient level where possible.
8. Cross-split leakage and source-bias checks.
9. Medical/manual review of flagged samples and a stratified sample of accepted data.
10. Freeze and export the model-ready manifest with an audit report.

5. File Integrity and Manifest Checks

Check	Classification	Segmentation	Action on failure
File opens successfully	Required	Required	REJECT if unrecoverable
Manifest path exists	Required	Required	REVIEW then fix manifest
Unique sample_id	Required	Required	Fix before any split
subject_id consistency	Strongly required	Strongly required	REVIEW; infer only with documented method
Image dimensions valid	Required	Required	REJECT zero/corrupt; flag extreme outliers
Mask path exists	N/A	Required	REJECT or REVIEW
Image-mask dimensions match	N/A	Required for 2D; geometry must match for 3D	REJECT until corrected
Allowed mask values only	N/A	Required	REVIEW/REJECT
Class in approved vocabulary	Required	If subtype label used	REVIEW mapping

6. Duplicate, Near-Duplicate, and Leakage Validation

Duplicate control is mandatory because medical image datasets often contain repeated slices, copies with different filenames, or samples reused across source datasets.

- **Exact duplicates:** compute cryptographic hashes for identical files.
- **Visual near-duplicates:** use perceptual hashing and/or image similarity to detect resized, recompressed, cropped, or slightly modified copies.
- **Patient leakage:** all slices, modalities, scans, and derived images from one subject must remain in one split.
- **Cross-source overlap:** compare newly collected sources against the existing project corpus, not only within each source.
- **Mask-derived leakage:** image and mask pairs from one patient must never be split across train/validation/test.
- **Original split audit:** do not blindly trust a source train/test split; verify for duplicate or subject overlap whenever IDs or similarity checks make this possible.

Split rule

Create model splits only after duplicate resolution and patient grouping. Prefer patient-level stratified splitting; image-level random splitting is not acceptable when multiple images/slices come from the same patient.

7. Classification Label Validation

Requirement	Method	Minimum acceptance
Approved classes	Map all source labels to glioma / meningioma / pituitary / no tumor	No unreviewed ambiguous labels in final manifest
Label provenance	Record whether labels are radiology/pathology/expert/source-folder derived	Provenance known at source level
Stratified manual review	Review random samples from every class and every source	At least 50 samples per class per source when feasible, or 10% for small sources
High-risk cases	Review borderline/atypical images, contradictory filenames/metadata, and label outliers	100% of flagged cases reviewed
Cross-source consistency	Check whether the same class visually differs systematically by source/preprocessing	Document source-specific differences and bias risk
No-tumor integrity	Ensure control images truly lack labeled tumor and are not obvious non-brain/non-MRI images	Flag uncertain controls for medical review

8. Segmentation Mask Validation

Check	Expected condition	Flag examples
Image-mask pairing	Exactly one correct mask set per image/volume	Missing mask, wrong subject, naming mismatch
Geometry	Same 2D shape; for 3D same volume geometry/orientation/spacing or a documented valid transform	Shifted, flipped, different slice count
Mask values	Integer IDs matching documented label map	Unexpected grayscale/anti-aliased values
Tumor presence	Tumor-positive case has non-empty tumor mask	Empty mask on tumor-positive sample
No-tumor case	Mask absent or correctly empty according to project convention	Non-empty tumor region in no-tumor case
Anatomical plausibility	Mask lies within the brain and follows visible lesion boundaries	Mask outside skull/brain, huge rectangle, text region
Connected components	Number/size of components clinically plausible	Thousands of speckles, single-pixel noise
Slice continuity for 3D	Tumor changes reasonably across adjacent slices	Abrupt isolated full-slice masks
Tumor size distribution	Area/volume ratios fall within plausible dataset distribution	Extreme outliers reviewed manually

For multi-region masks, preserve the source label scheme and validate each region separately. If the model will use a binary mask, collapse labels only in a reproducible derived dataset after the original masks pass validation.

9. Image Quality Validation

Quality issue	Detection approach	Disposition
Corrupt/unreadable image	Automated file read	Reject
Very low resolution / thumbnail	Dimension statistics + manual review	Reject or quarantine
Severe motion/artifact	Manual/medical review; automated outlier flagging where feasible	Review; reject if anatomy/lesion unusable
Excessive crop / missing brain	Brain coverage checks + manual review	Reject severe cases
Burned-in text / borders	Visual review and pixel-pattern checks	Review; remove only through documented preprocessing if safe
Non-MRI / wrong anatomy	Manual review / metadata	Reject
Preprocessing inconsistency	Source analysis (skull-stripped vs non-skull-stripped, contrast, background)	Do not necessarily reject; flag source bias and standardize later
Orientation inconsistency	Metadata/visual check	Standardize only in reproducible preprocessing

10. Dataset-Level Analysis Requirements

- Counts by class, source, subject, MRI sequence, and split.
- Class imbalance ratios before augmentation.

- Image width/height distribution; for 3D, volume shape and voxel spacing distributions.
- Source-specific preprocessing differences and scanner/institution differences where metadata allow.
- Tumor mask area/volume distribution overall and by tumor class/source.
- Percentage of empty masks, missing labels, missing subject IDs, and unknown modalities.
- Duplicate and near-duplicate counts: within source, across sources, and across final splits.
- Source contribution to each final split; avoid test sets dominated by a single source unless intentionally used as external validation.
- Representative contact sheets/overlays for each class and segmentation source for human inspection.

11. Split Strategy Requirements

Dataset purpose	Recommended split	Rules
Primary development	70% train / 15% validation / 15% test	Patient-level, stratified by tumor class where possible; keep each patient in one split
If data are limited	80% train / 10% validation / 10% test	Still patient-level; consider cross-validation only inside training/development data
External validation	Completely separate source/institution if possible	No overlap with training sources/patients; do not use for tuning
Segmentation	Same patient-level rule	All slices/modalities/masks for one patient stay together

Do not optimize on the test set

Once the final test set is frozen, it should not be repeatedly inspected to guide model decisions. Use validation data for tuning and reserve test data for final evaluation.

12. Medical/Manual Review Protocol

- A medically knowledgeable reviewer should evaluate all ambiguous labels/masks and a stratified sample of otherwise accepted data.
- Review must be blinded to model predictions during dataset validation to avoid circular confirmation.
- For every reviewed item, record reviewer decision, reason, confidence, and whether a second review is needed.
- Disagreements should be escalated to a more experienced reviewer or consensus review rather than silently resolved by the data team.
- If ground truth quality is uncertain, retain an uncertainty flag; do not force every case into a confident label.

13. Validation Output Manifest

Required column	Purpose
sample_id / subject_id	Stable identity and patient grouping
final_class_label	Approved project label after mapping/review

Required column	Purpose
validation_status	ACCEPTED / REVIEW / REJECTED / QUARANTINED
validation_reason	Short standardized reason code + notes
duplicate_group_id	Links exact/near-duplicate copies
quality_flags	motion, crop, text, low_resolution, source_bias, etc.
mask_status	valid / invalid / review / not_applicable
reviewer_status	not_reviewed / reviewed / consensus
final_split	train / validation / test / external / excluded
source_dataset	Preserve provenance
modality	T1/T1ce/T2/FLAIR/unknown
image_path + mask_path	Paths to validated, model-ready derived data

14. Model-Ready Acceptance Criteria

Check	Requirement
☐	All files in the final manifest are readable and structurally valid.
☐	No unresolved exact duplicate appears across train/validation/test.
☐	No known patient/subject appears in more than one split.
☐	Near-duplicate leakage review is complete.
☐	Every classification sample has an approved final label.
☐	Every segmentation sample has a validated image-mask pair and valid mask values.
☐	All REVIEW/REJECTED/QUARANTINED samples are excluded from model training unless their status is resolved.
☐	Class, source, modality, dimensions, and tumor-size analyses have been completed and documented.
☐	Split counts can be reproduced from the final manifest.
☐	A medical/manual review record exists for flagged data and for the required stratified audit sample.
☐	A frozen dataset version number and checksum/snapshot is created before model training begins.

15. Required Deliverables to the AI Model Team

Deliverable	Format	Must contain
Validated Master Manifest	CSV/XLSX	Final labels, status, flags, subject grouping, final split, paths, source
Validation Report	PDF/Markdown	Methods, rejection reasons, duplicate/leakage results, label/mask audit, limitations
Dataset Analysis Workbook	XLSX/CSV	Counts, class/source distributions, dimensions, modality, missingness, tumor-size statistics
Model-Ready Dataset	Files + manifest	Only ACCEPTED data arranged reproducibly; original raw data preserved separately
Manual Review Log	XLSX/CSV	Reviewer decision, reason, confidence, escalation/consensus
Exception/Exclusion Log	XLSX/CSV	All rejected/quarantined samples with reason
Dataset Version Record	TXT/JSON/README	Version ID, creation date, checksums or snapshot reference, split seed/method

16. Minimum Analysis Plots / Tables

- Bar chart/table: samples and unique subjects per class.
- Bar chart/table: samples per source and source composition per class.
- Histogram/table: image dimensions; for 3D, voxel spacing and volume shapes.
- Histogram/box summary: tumor mask area/volume ratios overall and by class/source.
- Table: exact duplicates and near-duplicate groups by split/source.
- Table: rejected/reviewed samples by reason.
- Cross-tab: class x final split and source x final split.
- Representative image grid with mask overlays for each tumor class/source.

17. Final Sign-off Questions

Check	Requirement
☐	Can we prove where every final sample came from?
☐	Can we prove that the same patient is not in multiple splits?
☐	Can we explain how every label and mask was obtained and validated?
☐	Are the final test cases independent enough to measure generalization rather than memorization?
☐	Could the model learn source/scanner/preprocessing shortcuts instead of tumor features? If yes, is that risk documented and mitigated?
☐	Can another team member reproduce the final dataset from the raw staging data + manifests + validation scripts/rules?
☐	Are limitations and uncertain labels/masks explicitly recorded instead of hidden?